



LAADAN-AC: Beyond Survival in Admissible Offline Treatment-Policy Learning

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ABSTRACT

What makes an offline treatment policy reliable? High estimated return alone does not show whether a learned policy respects explicit action constraints or remains aligned with expert behaviour. We study this distinction with Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), which embeds benchmark-defined admissibility directly in an offline actor interface. Using predetermined epoch-1000 checkpoints across five random seeds on Intensive Care Unit Sepsis (ICU-Sepsis), LAADAN-AC achieves 0.7927 ± 0.0012 expected return, 0.0000% selected-action inadmissibility, and 0.9540 ± 0.0032 expert argmax agreement. Against post-hoc masked Vanilla Offline Actor-Critic (VOAC), paired evaluation yields $+0.0275 \pm 0.0021$ return and $+0.2568 \pm 0.0022$ expert-agreement differences, although both are admissible after masking. Ablations identify hard masking as the direct feasibility mechanism, while the remaining terms shape the return-alignment operating point. An exploratory eICU Collaborative Research Database (eICU-CRD) Demo study further tests cross-source portability. Code and results are available at: <https://github.com/AnnyaB/laadan-ac>

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1. INTRODUCTION

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection and remains a major source of mortality and critical-care burden worldwide [1], [2]. Its treatment is fundamentally sequential: intravenous fluids, vasopressors, and other interventions must be adapted as physiology evolves, while the consequences of an inappropriate decision may compound over time [3]. Reinforcement learning (RL) offers a natural framework for this setting by learning treatment policies that map patient states to sequential actions from retrospective intensive-care data [4]. In clinical practice, however, online exploration is unacceptable. Policy learning must therefore proceed from fixed observational data, placing the problem in the offline RL setting. This changes what it means for a treatment policy to be reliable. A policy may achieve

a high estimated return while relying on state–action pairs that are weakly supported by the data from which it was learned. The resulting distribution shift can produce extrapolation error and optimistic value estimates for actions whose consequences are poorly identified [5], [6]. Early sepsis RL studies demonstrated the promise of data-driven sequential treatment optimisation [7], [8], yet subsequent work also revealed how strongly observational evaluation can depend on state construction, confounding, and off-policy estimation [9]. The challenge is therefore not only to learn which actions appear valuable, but also to determine the domain within which the learned policy should be allowed to act.

General offline RL addresses one part of this problem by controlling policy improvement relative to the observed data. Batch-Constrained Q-Learning restricts candidate actions toward regions supported by the dataset [10]. Conservative Q-Learning (CQL) suppresses optimistic value estimates under distribution shift [11], while Implicit Q-Learning (IQL) avoids direct maximisation over unseen actions during value learning [12]. These approaches reduce a central pathology of offline control, but they do not make data support and explicit feasibility equivalent. An action may be statistically plausible yet excluded by a specified admissibility rule; conversely, an admissible action need not be the action preferred by imitation or pessimistic value estimation. A policy can therefore be conservative without possessing an explicit notion of where action selection is permitted.

Recent clinical offline RL makes this distinction increasingly visible. Uncertainty-aware methods estimate confidence around treatment recommendations [13], while continuous-action formulations model dosing at finer resolution than discrete treatment bins [14]. Safe offline approaches have further addressed sparse rewards and variable episode lengths [15], action imbalance in treatment optimisation [16], and batch-constrained sepsis control [17]. More recent work introduces safety at different points in the learning pipeline. Offline Guarded Safe Reinforcement Learning (OGSRL) constrains policy improvement through an out-of-distribution (OOD) guardian and physiological safety-cost structure [18]. Safe Conservative Q-Learning (SafeCQL) combines conservative value regularisation with clinician-informed constraints for vasopressor optimisation [19]. Clinically-aware OOD-regularized Model-based Policy Optimization (CORMPO) combines clinically informed regularisation with model-based policy optimisation through a probabilistic digital twin [20]. Other work targets orthogonal sources of difficulty: Multimodal Offline REinforcement learning for Clinical notes Leveraged Enhanced stAte Representation (MORE-CLEAR) enriches patient-state representations using clinical notes and structured electronic health records [21], while Treatment Stitching (TreatStitch) augments clinically plausible trajectories to mitigate data sparsity [22]. Taken together, these methods suggest that reliability in offline treatment learning is not a single objective. It can arise through value pessimism, uncertainty estimation, trajectory constraints, representation learning, model-based reasoning, or improved coverage of the observed data. A different line of work makes the feasible action set itself part of the decision process. Invalid-action masking removes prohibited actions from the policy distribution before selection [23], while Pure-Past Action Masking (PPAM) formalises masking under history-dependent temporal constraints [24]. In clinical control, a recent two-stage sepsis framework similarly separates constraint-aware learning from runtime safety filtering, showing that feasibility can also be imposed after a policy has already been trained [25]. This distinction becomes particularly sharp in ICU-Sepsis, which provides a finite tabular Markov decision process (MDP) together with state-dependent admissible-action sets derived from empirical treatment support [26]. Here, admissibility is a property of the benchmark interface. It should not be conflated with prospective clinical safety, treatment optimality, or readiness for deployment.

This leads to the central question of our study: what changes when benchmark-defined admissibility becomes part of the actor while the policy is being learned, rather than a filter applied only after learning? Conservative methods primarily alter value estimation or proximity to behavioural support; guarded and constrained methods regulate broader state–action behaviour; runtime filters constrain the actions exposed by a trained policy. These mechanisms are closely related, but they intervene at different stages of policy formation. The distinction matters especially in healthcare, where candidate policies cannot ordinarily be compared through online deployment and model selection is itself a methodological problem [27]. Studying an admissibility-aware actor therefore requires more than reporting a single return: it requires separating the feasible action interface from the preferences learned within that interface.

We study this problem through Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), an offline actor-critic framework in which the benchmark admissibility mask is embedded directly in the actor during optimisation and action extraction. Within this interface, LAADAN-AC combines twin reward critics, conservative critic regularisation [11], expert-policy Kullback–Leibler (KL) regularisation [28], a state–action smoothness proxy [29], and adaptive Lagrangian cost pressure for constrained optimisation [30], [31]. The individual mechanisms are established. The contribution lies in organising them around an admissibility-constrained actor and in using that construction to separate two roles that are often treated together: defining which actions remain available to the policy and shaping which policy is learned within that feasible set. This separation gives LAADAN-AC a specific interpretation. The hard admissibility interface determines the support over which action selection can occur. The remaining objective terms act within that support: conservative regularisation shapes value estimates, expert-policy regularisation biases the learned distribution toward benchmark expert behaviour, the smoothness term discourages abrupt departures from nearby expert-supported structure, and the Lagrangian term supplies adaptive cost pressure during optimisation. The framework is therefore not designed around the assumption that every component should independently increase the same scalar measure. Rather, it provides a controlled setting in which feasibility, return, and expert alignment can be studied as distinct but coupled properties of an offline policy.

We evaluate this question on ICU-Sepsis under a predetermined fixed-final-epoch protocol. Behaviour Cloning (BC), CQL-regularised fitted-Q, Vanilla Offline Actor-Critic (VOAC), and LAADAN-AC are trained for 1,000 epochs with frozen hyperparameters and the same five random seeds {42, 43, 44, 45, 46}. The finite-horizon benchmark evaluator is not used for checkpoint selection, early stopping, seed selection, or hyperparameter adaptation; each policy is evaluated at the predetermined epoch-1000 checkpoint. To isolate the effect of learning within the admissible set, we also apply the benchmark mask post hoc to the same final VOAC checkpoints, without retraining. Reported intervals are two-sided 95% t -based confidence intervals across the five training seeds and quantify variation across optimisation initialisations rather than patient- or population-level uncertainty. We additionally construct an eICU Collaborative Research Database (eICU-CRD) Demo MDP for exploratory cross-source evaluation [32], [33].

Our contributions are:

- We formulate admissible offline treatment-policy learning as an actor-interface problem, separating policies learned within a benchmark-defined feasible set from policies constrained only at action extraction.

- We propose Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), which embeds this interface directly in policy optimisation while combining conservative value regularisation, expert-policy shaping, state–action smoothness, and adaptive Lagrangian cost control.
- We compare training-time admissibility with post-hoc filtering using the same five final VOAC checkpoints, testing whether the same admissibility rule produces different policies when imposed during rather than after optimisation.
- We analyse the mechanism at component and diagnostic levels, using controlled ablations together with state-, action-, value-, and trajectory-level evidence to distinguish feasible-support enforcement from policy shaping.
- We test cross-source portability through a separately constructed eICU-CRD Demo MDP, examining whether the same actor-interface behaviour persists under a second critical-care data source.

The study therefore asks a narrow question: does a policy change when admissibility is part of learning rather than a final action filter? Section 2 formalises this question and the evaluation protocol; Section 3 reports the return–admissibility results, ablations, admissibility-failure analysis, and cross-source study; Section 4 concludes the paper.

2. METHODS AND MATERIAL

2.1 Problem formulation

We formulate offline treatment-policy learning as a finite-horizon constrained Markov decision process (CMDP), following the standard formulation for sequential decision problems with reward objectives and cost constraints [30]. The CMDP is defined as:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \mathcal{C}, \gamma, H) \quad (1)$$

where $\mathcal{S}$ is the finite state space, $\mathcal{A}$ is the discrete treatment-action space, $\mathcal{P}(s' | s, a)$ is the transition probability from state s to the next state s' after action a , $\mathcal{R}(s, a)$ is the expected immediate reward, $\mathcal{C}(s, a)$ is the inadmissibility cost, γ is the discount factor, and H is the finite evaluation horizon. A policy $\pi(a|s)$ maps each patient state to a distribution over treatment actions. We train all policies offline from fixed benchmark files and evaluate them without online interaction.

The expected immediate reward is defined from the reward assigned to the next state:

$$\mathcal{R}(s, a) = \sum_{s' \in \mathcal{S}} \mathcal{P}(s' | s, a) r(s') \quad (2)$$

where $r(s')$ is the next-state reward: $r(s') = 1$ for the survival terminal state and $r(s') = 0$ for death and intermediate states. For each state, we define the admissible action set as:

$$\mathcal{A}_{adm}(s) = \{a \in \mathcal{A}: m(s, a) = 1\} \quad (3)$$

where $m(s, a)$ is a binary benchmark admissibility mask. Here, $m(s, a) = 1$ means that action a is admissible in state s , and $m(s, a) = 0$ means that it is inadmissible. The immediate inadmissibility cost was defined as:

$$\mathcal{C}(s, a) = 1 - m(s, a) \quad (4)$$

Thus, admissible actions have cost 0, and inadmissible actions have cost 1. This definition treats admissibility as a benchmark-defined action-support constraint.

2.2 Data sources and MDP construction

We use the ICU-Sepsis benchmark as the primary tabular MDP for evaluating sepsis treatment policies [26]. The benchmark contains 716 states, 25 discrete treatment-action bins, 47-dimensional standardised state-centre features, an initial-state distribution, transition probabilities, next-state rewards, a benchmark expert policy and an admissible-action mask. The terminal states represent death and survival. All policies were evaluated with $H = 50$ and $\gamma = 1.0$. For policy regularisation and alignment analysis, we filter the released expert distribution through the benchmark admissibility mask and renormalise it; we denote the resulting benchmark reference distribution by π_E .

To test whether the same pipeline transfers beyond the released benchmark, we construct a second MDP from the eICU Collaborative Research Database Demo v2.0.1 [32], [33]. The eICU-CRD Demo is a real, de-identified intensive care dataset, but it is not distributed as an offline reinforcement learning environment. We therefore wrote an adapter that converts the eICU Demo SQLite database into the same set of file interfaces used by ICU-Sepsis: `transitionFunction.csv`, `rewardFunction.csv`, `initialStateDistribution.csv`, `expertPolicy.csv`, `admissibleActions.txt`, and `stateClusterCenters.csv`.

The adapter formed hourly ICU records from vital signs, selected laboratory variables and patient metadata. These records were standardised and clustered into 200 non-terminal states using MiniBatchKMeans. Two absorbing terminal states were added for death and survival/discharge, bringing the total to 202 states. Treatment records from medication, infusion, treatment and respiratory-care tables were mapped into a 25-action grid. The grid used two treatment axes: haemodynamic support and anti-infective/respiratory support. This kept the action-space size consistent with ICU-Sepsis while using only treatment events observed in the eICU Demo.

The empirical clinician action distribution in each state cluster was used as the benchmark expert policy for the eICU Demo MDP. The admissibility mask was derived from empirical state-action support. For each eICU state cluster, actions with at least two observed occurrences were marked admissible. If fewer than two actions met this threshold, the two most frequent observed actions were retained where available. State rows with no observed action support were left unconstrained as a validity fallback. Transition probabilities were estimated from observed patient-hour transitions. The resulting eICU Demo MDP contained 2,309 ICU stays, 76,051 observed transitions, 22-dimensional state features, 25 actions, 200 non-terminal states and two

terminal states. This experiment was used as a small real-data cross-domain portability check, not as a full external clinical validation.

Table 1 summarises the two data sources and clarifies the difference between the released ICU-Sepsis benchmark and the eICU-CRD Demo MDP constructed in this study.

Table 1: Benchmark and portability-task characteristics

Property	ICU-Sepsis	eICU-CRD
Source	Released ICU-Sepsis benchmark	eICU-CRD Demo v2.0.1 converted into MDP
Task purpose	Main benchmark	Exploratory cross-source portability
Number of states	716	202
Non-terminal states	714	200
Number of actions	25	25
Feature dimension	47	22
Terminal states	Survival: 714, death: 713	survival/discharge: 201, death: 200
Horizon	50	50
Reward definition	Survival: 1, death/intermediate: 0	Survival/discharge: 1, death/intermediate: 0
Admissibility definition	Released benchmark admissibility mask	Empirical state-action support mask
Bellman transition terms per pass	$716 \times 716 \times 25 = 12,816,400$	$202 \times 202 \times 25 = 1,020,100$

Note: *ICU* = intensive care unit; *eICU-CRD* = eICU Collaborative Research Database; *MDP* = Markov decision process. **Bellman transition terms per pass** were calculated as the number of states $\times$ number of states $\times$ number of actions.

2.3 LAADAN-AC framework

With the benchmark interface fixed, we turn to the policy-learning problem itself. We propose Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), an offline actor-critic framework in which benchmark-defined admissibility is part of the actor interface during optimisation and action extraction. LAADAN-AC organises established mechanisms—action masking, twin reward critics, conservative critic regularisation, expert-policy regularisation, a state–action smoothness proxy, and adaptive Lagrangian cost control—around this constrained interface. The contribution lies in this organisation and in the mechanism-level analysis it enables, rather than in treating the constituent mechanisms as individually novel.

Figure 1 shows the model flow. The fixed MDP supplies the state feature vector $x(s_t)$, transition matrix $\mathcal{P}$, reward table $\mathcal{R}$, admissibility mask m and admissibility-filtered benchmark expert policy π_E . The actor maps $x(s_t)$ to one logit for each of the 25 treatment-action bins. The admissibility mask is applied before action selection, so the deployed policy selects only from $A_{\text{adm}}(s)$. The critic maps the same state representation to two reward-value estimates, $Q_1(s, a)$ and $Q_2(s, a)$. LAADAN-AC adds a cost head, $\hat{C}(s, a)$, for benchmark-defined inadmissibility cost. Actor optimisation uses the lower reward estimate, $\min(Q_1(s, a), Q_2(s, a))$, to reduce value overestimation.

Vanilla Offline Actor-Critic (VOAC) and LAADAN-AC used separate actor and critic encoders. Each encoder was a two-layer multilayer perceptron with rectified linear units and a dropout rate of 0.10. For ICU-Sepsis, the input dimension was 47, the hidden and latent dimensions were 128, and the action dimension was 25. The online LAADAN-AC network contained 58,212 trainable scalar parameters. For the eICU-CRD Demo MDP, the input dimension was 22, and the action dimension remained 25. We initialise the target networks from their online counterparts and update them gradually toward the current parameters, providing stable Bellman targets without introducing additional parameters at deployment. The dominant Bellman computation scales as $O(|S|^2|A|)$, with benchmark-specific transition counts reported in Table 1.

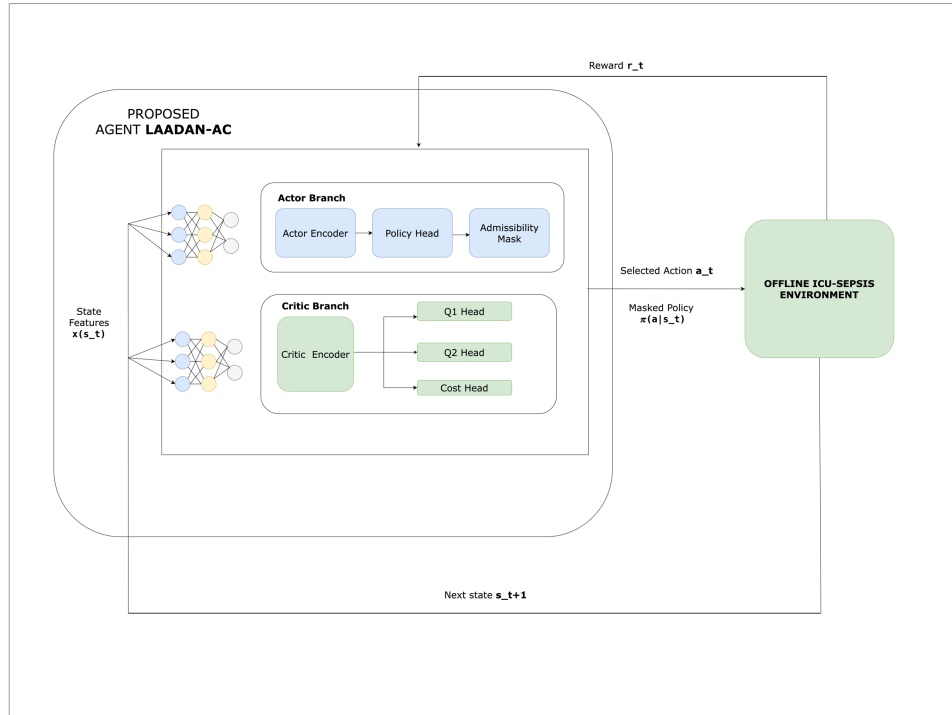


Fig. 1: LAADAN-AC Architecture Diagram. The fixed MDP provides state features, transition probabilities, rewards, admissibility masks and the benchmark expert policy. The actor produces treatment-action logits, the admissibility mask restricts action selection, twin reward critics estimate action values, and the cost head estimates a benchmark-defined inadmissibility cost for Lagrangian control

2.4 Training objective and admissibility mechanisms

LAADAN-AC separates two roles within the learning objective: a hard admissibility interface that defines the feasible policy support, and four soft terms that shape the policy learned within that support

A. HARD ADMISSIBILITY INTERFACE

First, it *masks* inadmissible actions before normalising the actor logits, following the standard invalid-action masking principle [23]:

$$\pi(a | s) = \frac{\exp(\ell(s, a))m(s, a)}{\sum_{a' \in \mathcal{A}} \exp(\ell(s, a'))m(s, a')} \quad (5)$$

where $\pi(a|s)$ is the learned policy, $\ell(s, a)$ is the actor logit for action a in state s , $\mathcal{A}$ is the discrete action set, and $m(s, a)$ is the binary admissibility mask. Here, $m(s, a) = 1$ permits action a and $m(s, a) = 0$ removes it from the policy. In implementation, inadmissible logits were assigned a large negative value before softmax. Because every benchmark state is assigned at least one admissible action, masked greedy extraction selects exclusively from $\mathcal{A}_{\text{adm}}(s)$. This is the mechanism that directly enforces *benchmark-defined* selected-action admissibility.

B. POLICY SHAPING WITHIN THE FEASIBLE SET

Second, the critic used a *conservative value penalty* derived from Conservative Q-Learning (CQL) [11]. This term appears in two places. The CQL-regularised fitted-Q model is a separate baseline. LAADAN-AC uses the same conservative idea only as one term in its reward-critic loss:

$$L_{\text{CQL}} = \alpha \mathbb{E}_{s \sim D} [\log \sum_{a \in \mathcal{B}(s)} \exp(Q(s, a)) - \sum_{a \in \mathcal{A}} \pi_E(a | s) Q(s, a)] \quad (6)$$

where L_{CQL} is the conservative critic penalty, α is its weight, D is the offline state distribution, $Q(s, a)$ is the learned action-value estimate, and $\pi_E(a | s)$ is the benchmark expert-safe policy. For the CQL-regularised fitted-Q baseline, $\mathcal{B}(s) = \mathcal{A}$. For LAADAN-AC, $\mathcal{B}(s) = \mathcal{A}_{\text{adm}}(s)$, so conservative regularisation is evaluated over the same feasible support available to the actor. CQL therefore serves as the conservative value-learning comparator, whereas LAADAN-AC uses the CQL-derived penalty as one term within the actor-critic objective.

Third, LAADAN-AC regularised the actor toward the *benchmark expert-safe policy* using the KL divergence [28]:

$$L_{\text{KL}} = \mathbb{E}_{s \sim D} [DKL(\pi_E(\cdot | s) \parallel \pi(\cdot | s))] \quad (7)$$

where L_{KL} is the expert-policy regularisation term, D is the offline state distribution, π_E is the benchmark expert-safe policy and π is the learned policy. The expert policy is a benchmark reference distribution.

The fourth component was a benchmark-specific *state-action smoothness proxy*. This term is not a direct implementation of continuous-control action-smoothing methods, such as Conditioning for Action Policy Smoothness (CAPS) [29]. Instead, it adapts the broader idea of action regularisation to the discrete treatment-action grid by penalising actions far from the expert-supported mean action index. For each state, the expert-supported mean action index and the smoothness cost were defined as

$$\bar{a}_E(s) = \sum_{a \in \mathcal{A}} a \pi_E(a | s), \quad S(s, a) = \frac{|a - \bar{a}_E(s)|}{|\mathcal{A}| - 1} \quad (8)$$

where $\bar{a}_E(s)$ is the expert-supported mean action index in the state s , $S(s, a)$ is the benchmark-specific smoothness proxy, and $|\mathcal{A}|$ is the number of actions. This term encourages the learned actor to avoid large deviations from the expert-supported action region, but it does not enforce temporal smoothness between consecutive treatment decisions.

Fifth, inadmissibility was treated as a *cost constraint*. Following the CMDP formulation [30], we impose the expected-cost objective:

$$\max_{\pi} E_{\pi}[\mathcal{R}] \quad \text{subject to} \quad E_{\pi}[\mathcal{C}] \leq d \quad (9)$$

where $E_{\pi}[\mathcal{R}]$ is the *expected return under the policy* π , $E_{\pi}[\mathcal{C}]$ is the expected inadmissibility cost, and d is the cost budget. We optimise the corresponding non-negative Lagrange multiplier using projected ascent [30], [31]:

$$\lambda \leftarrow \max(0, \lambda + \rho(E[\mathcal{C}] - d)) \quad (10)$$

where λ is the Lagrange multiplier and ρ is the Lagrange learning rate. If the expected cost exceeded d , λ increased, and the actor received a larger cost penalty. If the expected cost was below d , λ decreased or remained at zero after projection. The Lagrange multiplier is used here as an adaptive penalty on the expected benchmark-defined inadmissibility cost. In the main LAADAN-AC setting, inadmissible actions are already removed from the actor distribution by the admissibility mask in Equation (5).

Because the hard mask already excludes inadmissible actions from the actor distribution, the Lagrangian term has a different role: it provides adaptive cost pressure during optimisation rather than defining the feasible action set itself. The no-mask component intervention in Section

3 tests whether the remaining objective can reproduce this feasibility property when the hard interface is removed.

The actor loss function combines these components:

$$L_{\text{actor}} = -\mathbb{E}a \sim \pi(\cdot | s)[Q(s, a)] - \beta H(\pi(\cdot | s)) + \lambda \mathbb{E}a \sim \pi(\cdot | s)[\hat{C}(s, a)] + \eta L_{\text{KL}} + \mu \mathbb{E}_{a \sim \pi(\cdot | s)}[S(s, a)] \quad (11)$$

where $Q(s, a) = \min(Q_1(s, a), Q_2(s, a))$, Q_1 and Q_2 are the twin reward critics, $H(\pi(\cdot | s))$ is policy entropy, β is the entropy coefficient, $\hat{C}(s, a)$ is the learned cost estimate, μ is the smoothness weight and η is the expert-policy regularisation weight.

We jointly optimise the masked actor, twin critics, conservative penalty, expert regularisation, smoothness proxy and Lagrangian cost term in Equations (5)–(11). Algorithm 1 summarises the resulting optimisation procedure and the predetermined final-checkpoint evaluation used in the confirmatory experiments.

Here, θ denotes the actor parameters, ϕ_1 and ϕ_2 denote the two reward-critic parameter sets, ψ denotes the cost-critic parameters, α is the conservative critic weight, η is the expert-policy regularisation weight, μ is the smoothness weight, β is the entropy coefficient, γ is the discount factor, τ is the target-network update coefficient, ρ is the Lagrange learning rate, d is the cost budget, E is the number of training epochs.

Algorithm 1: LAADAN-AC Training under a Predetermined Fixed Schedule

Require: Finite-horizon benchmark MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, \mathcal{R}, C, \gamma, H)$, state features x_s , admissibility mask $m(s, a)$, benchmark expert policy $\pi_E(a | s)$

Require: Actor π_θ , reward critics Q_{ϕ_1}, Q_{ϕ_2} , cost critic C_ψ , and target networks $\bar{\theta}, \bar{\phi}_1, \bar{\phi}_2, \bar{\psi}$

Require: Conservative weight α , expert-KL weight η , smoothness weight μ , entropy coefficient β , cost budget d , Lagrange step size ρ , target-update rate τ

Require: Predetermined training horizon $E = 1000$

1: Initialise $(\theta, \phi_1, \phi_2, \psi)$; set target parameters equal to online parameters

2: $\lambda \leftarrow 0$

3: // Phase 1: admissibility-constrained policy optimisation

4: for $e = 1$ to E do

5: $\bar{\pi}(\cdot | s) \leftarrow \text{MaskedSoftmax}(\bar{z}_{\bar{\theta}}, (s)m(s, \cdot))$

- 6:** Compute reward and cost Bellman targets using $P, \mathcal{R}, C, \bar{\pi}$, $\min(\bar{Q}_{\bar{\phi}_1}, \bar{Q}_{\bar{\phi}_2})$, entropy regularisation, and current cost pressure $\lambda \bar{C}_{\bar{\psi}}$
- 7:** Update (ϕ_1, ϕ_2, ψ) using masked reward- and cost-Bellman losses + $\alpha[\mathcal{R}_{\text{CQL}}(Q_{\phi_1}) + \mathcal{R}_{\text{CQL}}(Q_{\phi_2})]$
- 8:** $\pi_\theta(\cdot | s) \leftarrow \text{MaskedSoftmax}(z_\theta, (s)m(s, \cdot))$
- 9:** Compute

$$J_Q = \mathbb{E}_{a \sim \pi_\theta} [\min(Q_{\phi_1}, Q_{\phi_2})], \quad J_C = \mathbb{E}_{a \sim \pi_\theta} [C_\psi],$$

$$\mathcal{H}(\pi_\theta), \quad D_{\text{KL}}(\pi_E \parallel \pi_\theta), \quad J_{\text{smooth}}$$

- 10:** $\mathcal{L}_{\text{actor}} \leftarrow -J_Q - \beta \mathcal{H}(\pi_\theta) + \lambda J_C + \eta D_{\text{KL}}(\pi_E \parallel \pi_\theta) + \mu J_{\text{smooth}}$
- 11:** $\theta \leftarrow \theta - \nabla_\theta \mathcal{L}_{\text{actor}}$
- 12:** $(\bar{\theta}, \bar{\phi}_1, \bar{\phi}_2, \bar{\psi}) \leftarrow (1 - \tau)(\bar{\theta}, \bar{\phi}_1, \bar{\phi}_2, \bar{\psi}) + \tau(\theta, \phi_1, \phi_2, \psi)$
- 13:** $\lambda \leftarrow [\lambda + \rho(\mathbb{E}_s[J_C(s)] - d)]_+$
- 14: end for**
- 15: // Phase 2: predetermined final-checkpoint evaluation**
- 16:** $\pi_{\theta_E}^{\text{greedy}}(s) \leftarrow \arg \max_{a: m(s,a)=1} z_{\theta_E}(s, a)$
- 17:** $\mathcal{M}_E \leftarrow \text{ExactFiniteHorizonEval}(\pi_{\theta_E}^{\text{greedy}}, P, \mathcal{R}, C, H)$
- 18:** Save the predetermined epoch- E checkpoint and $\mathcal{M}_E$
- 19: return** $\theta_E, \mathcal{M}_E$

2.5 Baselines and ablations

We compare LAADAN-AC against baselines chosen to isolate imitation, conservative value learning and unrestricted actor-critic optimisation. Table 2 lists the models used in the comparison. The baselines were selected to isolate the main design choices relevant to offline treatment-policy learning: imitation from the benchmark expert-safe policy, conservative offline value learning and unconstrained actor-critic optimisation. BC learned a supervised policy from the benchmark expert-safe action distribution. It provides an imitation baseline that preserves expert alignment but does not directly optimise the MDP return. CQL was included through a CQL-

regularised fitted-Q baseline [11]. This model used Bellman regression with the conservative value penalty in Equation (6).

It was selected because conservative value learning directly addresses value overestimation under offline distribution shift, a central failure mode in unsupported-action selection. The implementation is therefore reported as a CQL-regularised fitted-Q baseline rather than as a full reproduction of the original CQL algorithm.

VOAC retains the shared actor-critic architecture while removing the admissibility interface and all LAADAN-AC policy-shaping terms, providing an unrestricted reference. To isolate learning under the admissibility interface from filtering only after learning, we also evaluate each final VOAC checkpoint with the same admissibility mask applied post hoc, without retraining or parameter updates [25]. We further perform six fixed-schedule ICU-Sepsis ablations—masking-only, no conservative critic, no expert KL, no smoothness proxy, no Lagrangian control, and no action mask—across the same five seeds; the full-model reference reuses the main LAADAN-AC checkpoints.

Table 2: Compared methods and role in the analysis

Method	Description	Role
BC	Supervised policy fitted to the benchmark expert distribution	Expert imitation
CQL-regularised fitted -Q	Fitted-Q model with Bellman regression and CQL-derived conservative regularisation	Conservative value learning
VOAC	Unrestricted actor-critic using the shared architecture without the LAADAN-AC admissibility and policy-shaping terms	Actor-critic reference
Post-hoc masked VOAC	Final VOAC checkpoint with the admissibility mask applied only at action extraction; no retraining	Training-time against post-hoc admissibility
LAADAN-AC	Actor-critic with hard admissibility masking, twin reward critics, conservative critic regularisation, expert KL, smoothness proxy, cost critic, and Lagrangian control	Full framework (Proposed method)
Component variants	Masking-only; no conservative critic; no expert KL; no smoothness; no Lagrangian; no action mask	Mechanism decomposition

Note: **BC** = Behaviour Cloning; **CQL** = Conservative Q-Learning; **VOAC** = Vanilla Offline Actor-Critic; **LAADAN-AC** = Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic.

2.6 Evaluation protocol

We evaluate the confirmatory comparison under a predetermined fixed-final-epoch protocol. Behaviour Cloning (BC), CQL-regularised fitted-Q, VOAC, and LAADAN-AC are trained for 1,000 epochs with frozen hyperparameters and the same five random seeds {42, 43, 44, 45, 46}. The

benchmark evaluator is not used for checkpoint selection, early stopping, seed selection, or hyperparameter adaptation. Instead, each run is evaluated at the pre-specified epoch-1000 checkpoint. The fixed-schedule wrapper enforces this separation by placing the evaluation interval beyond the training horizon and independently verifies that epoch 1000 is the only benchmark-evaluated checkpoint.

For each fixed checkpoint, primary policy extraction follows the model output. BC uses the greedy action from its policy logits; CQL-regularised fitted-Q uses the greedy action from learned Q-values; VOAC uses the greedy action from its unmasked actor logits; and LAADAN-AC uses masked greedy action selection. For the post-hoc comparison, we reuse the corresponding final VOAC checkpoint and apply the benchmark mask only at action extraction, without retraining or updating model parameters.

Primary metrics are computed by exact finite-horizon policy evaluation on the known transition matrices. Expected return follows Equation (2), while benchmark-defined inadmissibility is the expected cumulative inadmissibility cost divided by expected decision length. Expert argmax agreement compares the learned greedy action with the benchmark expert action; KL divergence measures full-distribution deviation from the expert reference distribution; mean action deviation uses the expert-supported mean action index in Equation (8); and policy entropy measures policy concentration. Monte Carlo trajectory sampling is reserved for diagnostic analyses and is not used to obtain the headline scores. All reported intervals are two-sided 95% t-based confidence intervals across the five random training seeds. They quantify variability across optimisation initialisations and do not represent uncertainty across patients, hospitals, or a clinical population. The same four trained methods are evaluated on the constructed eICU-CRD Demo MDP as an exploratory cross-source portability study [32], [33].

Table 3: Training and evaluation settings

Setting	BC	CQL-regularised fitted-Q	VOAC	LAADAN-AC
Optimiser	Adam	Adam	Adam	Adam
Weight decay	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
Gradient clipping	—	5.0	5.0	5.0
Seeds	42-46	42-46	42-46	42-46
Training epochs	1000	1000	1000	1000
Benchmark evaluation before epoch 1000	None	None	None	None
Reported checkpoint	Epoch 1000	Epoch 1000	Epoch 1000	Epoch 1000
Hidden dimension	128	128	128	128

Latent dimension	128	128	128	128
Dropout	0.10	0.10	0.10	0.10
Discount factor γ	—	1.0	1.0	1.0
Horizon H	50	50	50	50
Actor learning rate	0.001	—	0.001	0.0005
Critic learning rate	—	0.001	0.001	0.001
Entropy coefficient β	0	—	0.02	0.001
Conservative weight α	—	0.5	—	0.5
Expert KL weight η	—	—	—	0.005
Smoothness weight μ	—	—	—	0.001
Cost budget d	—	—	—	0.0
Lagrange learning rate ρ	—	—	—	0.0002
Target-update rate τ	—	0.02	0.02	0.01
Trainable parameters	25,881	25,881	54,987	58,212

Note: *BC, CQL, VOAC and LAADAN-AC are defined in Table 2. KL = Kullback-Leibler; α = conservative weight; η = expert-policy regularisation weight; μ = smoothness weight; ρ = Lagrange learning rate; τ = target-network update coefficient; γ = discount factor; d = cost budget; — = not applicable.*

The ICU-Sepsis component variants follow the same fixed epoch-1000 schedule and five-seed protocol. Post-hoc masked VOAC is evaluation-only. Safety-failure diagnostics aggregate the fixed final checkpoints across all five seeds, while the qualitative state-, action-, value-, and trajectory-level panels use a predeclared illustrative seed rather than one selected by performance.

3. RESULTS AND DISCUSSION

3.1 Results

We first compare the four independently trained policies under the predetermined epoch-1000 protocol (Table 4). LAADAN-AC attains the highest mean return (0.7927 ± 0.0012), zero selected-action inadmissibility, and the highest expert argmax agreement (0.9540 ± 0.0032). BC also yields zero greedy inadmissibility, CQL reduces but does not eliminate inadmissible selection, and VOAC departs most strongly from the admissible set.

Table 4: Fixed-final-epoch ICU-Sepsis performance

Model	Survival/return	Inadmissibility	Expert argmax match	KL to expert
BC	0.7834 ± 0.0013	0.0000 ± 0.0000	0.9533 ± 0.0038	3.9481 ± 0.0038
CQL-reg. fitted-Q	0.7761 ± 0.0024	0.0005 ± 0.0004	0.8545 ± 0.0126	4.1120 ± 0.0204
VOAC	0.7632 ± 0.0002	0.2357 ± 0.0009	0.0774 ± 0.0002	16.5103 ± 0.0014
LAADAN-AC	0.7927 ± 0.0012	0.0000 ± 0.0000	0.9540 ± 0.0032	3.9515 ± 0.0033

Note: Values are mean $\pm$ 95% confidence interval across *five random* training seeds. *Inadmissibility* is reported as a fraction; intervals describe training-seed variation.

Training under admissibility versus post-hoc masking. To test whether the admissibility interface must participate in learning, we apply the benchmark mask to each final VOAC checkpoint only at action extraction, without retraining. LAADAN-AC increases return ($+0.0275 \pm 0.0021$) and expert argmax agreement ($+0.2568 \pm 0.0022$), while reducing KL divergence (-1.0871 ± 0.0218) and mean action deviation (-1.2571 ± 0.0192). Both policies achieve zero selected-action inadmissibility. This shows that applying the same feasible action set after training does not recover the policy learned when admissibility constrains optimisation itself.

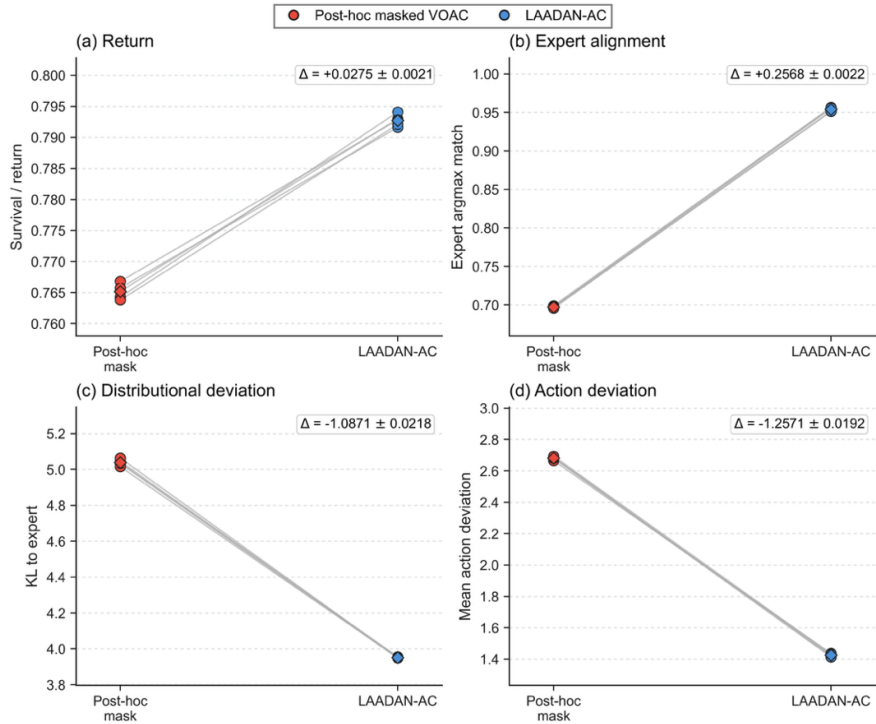


Fig. 2: Training under admissibility versus post-hoc masking. Lines connect the two policies from the same seed across the four reported metrics

Component ablation. We next remove each LAADAN-AC component under the same fixed schedule (Figure 3). Masking-only reaches 0.7974 ± 0.0026 return and the variant without conservative critic regularisation reaches 0.8067 ± 0.0009 , both with zero inadmissibility, while expert agreement falls to 0.7488 ± 0.0035 and 0.7473 ± 0.0025 . Removing the mask reintroduces inadmissible selection, identifying hard masking as the direct feasibility mechanism.

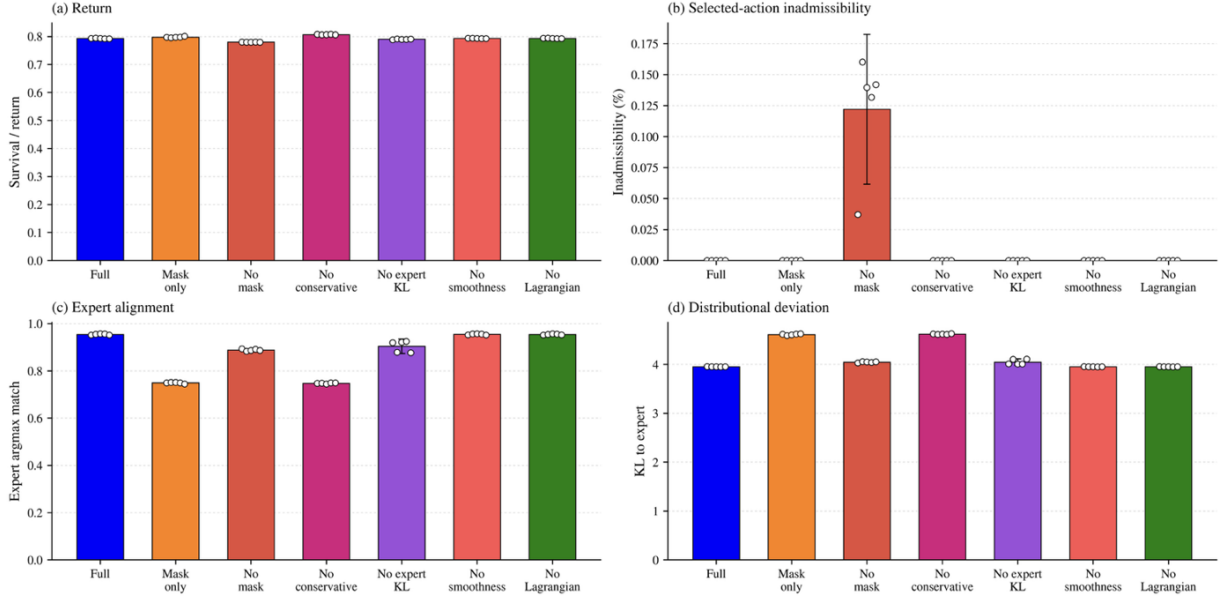


Fig. 3: Component decomposition of LAADAN-AC. Six fixed-schedule interventions isolate feasibility and policy-shaping effects

Cross-source portability. We evaluate the same fixed-schedule pipeline on the independently constructed eICU-CRD Demo MDP (Fig. 4), reporting expected return, selected-action inadmissibility, expert argmax agreement, and KL divergence to the benchmark reference distribution. CQL attains the highest mean return (0.7351 ± 0.0016), while LAADAN-AC obtains 0.7329 ± 0.0008 return with zero selected-action inadmissibility and 0.9984 ± 0.0006 expert agreement. Unrestricted VOAC deteriorates to 0.0447 ± 0.0322 return and $78.04\% \pm 4.70\%$ selected-action inadmissibility. Applying the admissibility mask post hoc removes selected-action inadmissibility from VOAC but does not recover its return or expert-alignment profile. The same actor-interface distinction observed on ICU-Sepsis therefore persists under this separately constructed MDP. We interpret this experiment as exploratory cross-source portability rather than external clinical validation.

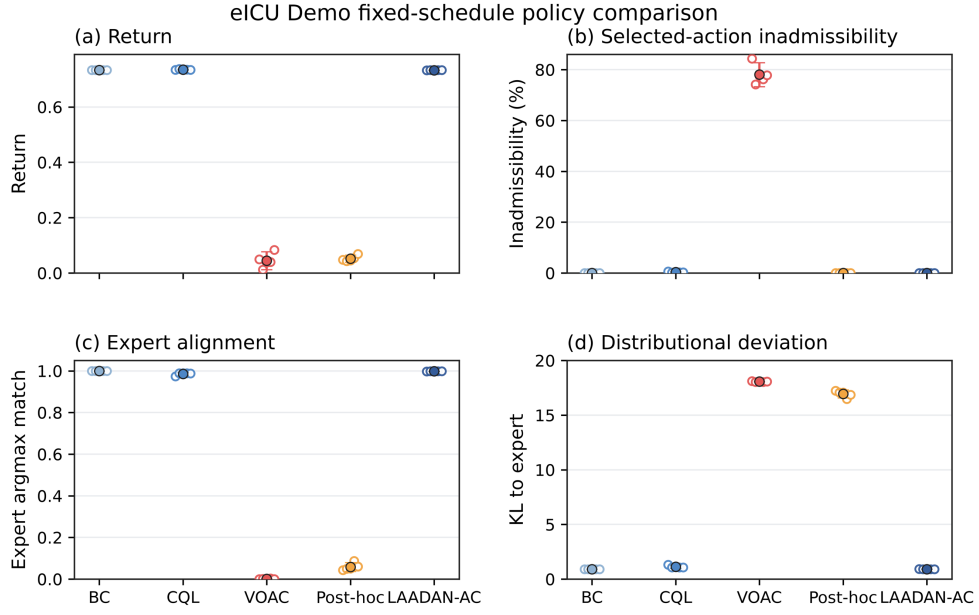


Fig. 4: Detailed fixed-schedule comparison on the eICU-CRD Demo MDP. The panels report expected return, selected-action inadmissibility, expert argmax agreement, and KL divergence to the benchmark reference distribution. Open markers show individual training seeds; central markers and error bars show five-seed means and 95% t -based confidence intervals across training seeds. All methods are evaluated at the predetermined epoch-1000 checkpoint

Admissibility-failure analysis. Finally, Figure 5 localises failure beyond aggregate rates. The five-seed state summary is complemented by action-level violations, Q-value separation, and a trajectory from the same initial state from predeclared seed 42, showing where unrestricted VOAC leaves the admissible set and how masked selection blocks that route.

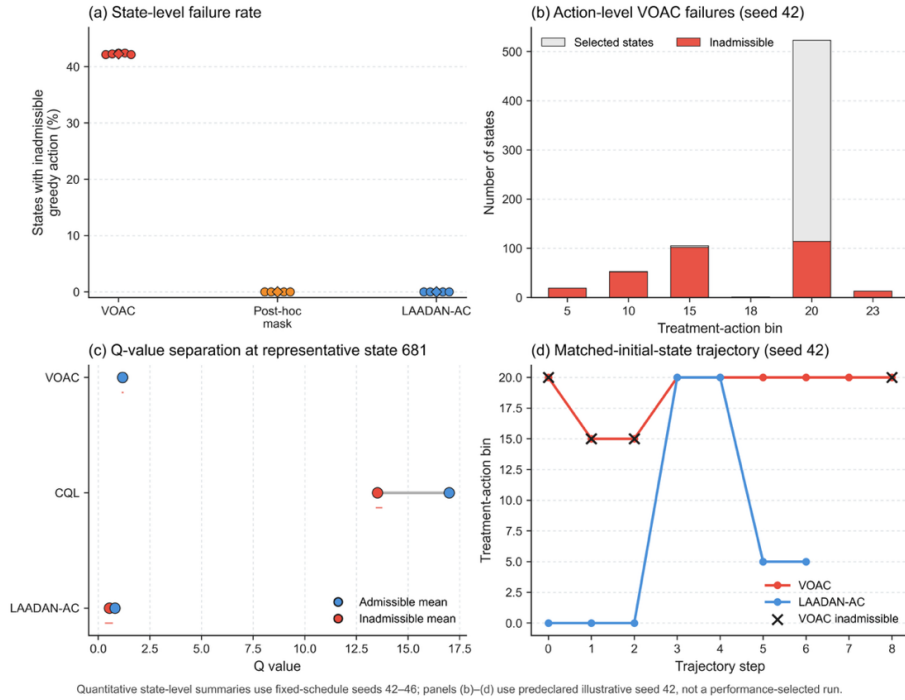


Fig. 5: Admissibility-failure diagnostics. State-, action-, value-, and trajectory-level views localise inadmissible behaviour

3.2 Discussion

The results identify where the observed behaviour comes from. VOAC optimises the benchmark return over the full action space, so actions outside the admissible set can still receive high value and be selected. CQL reduces this failure by penalising unsupported values, but its final greedy action is still chosen from the unrestricted set; conservative value learning therefore lowers inadmissibility without enforcing feasibility. BC follows the benchmark reference policy closely and consequently has zero greedy inadmissibility, but this behaviour comes from imitation rather than an explicit action constraint. LAADAN-AC changes the decision rule itself: the actor selects only from the admissible set, while reward optimisation proceeds inside that restricted support. This explains the combination of zero selected-action inadmissibility, high return, and strong expert agreement observed on ICU-Sepsis.

The post-hoc experiment separates action filtering from policy learning. Applying the same mask to a trained VOAC policy removes inadmissible selections, yet it does not recover LAADAN-AC's return, expert agreement, KL divergence, or action-deviation profile. We interpret this difference as an optimisation effect. In LAADAN-AC, inadmissible actions are absent from the actor's decision space throughout training, so gradients shape preferences only among feasible actions. Post-hoc masking instead removes actions after an unrestricted policy has already been learned. The two procedures therefore satisfy the same final admissibility rule while producing different policies.

The ablation study further separates the roles of the hard and soft components. Removing the mask reintroduces inadmissible action selection, identifying masked action selection as the direct mechanism responsible for feasibility. In contrast, masking-only and the variant without conservative critic regularisation achieve higher return than the full model while losing substantial expert agreement. The full objective therefore selects a different return–alignment operating point inside the admissible set; it is not the return-maximising configuration. The small change observed when Lagrangian control is removed under an active mask is also consistent with this mechanism: once selected actions are already restricted to zero benchmark-defined inadmissibility, little selected-action cost remains for the Lagrangian term to change. The remaining regularisers primarily influence which feasible actions the actor prefers.

The eICU experiment tests whether this structure persists under a separately constructed MDP. CQL attains a slightly higher mean return, whereas LAADAN-AC preserves zero inadmissibility and near-complete expert agreement. VOAC deteriorates sharply in both return and inadmissibility. We therefore read the eICU result as evidence that the actor-interface mechanism transfers across these two benchmark constructions, not as external clinical validation.

Finally, the diagnostic analysis shows how the aggregate failure arises. VOAC violations appear across states and treatment-action bins, and the representative Q-value and trajectory views show how inadmissible actions can remain attractive to an unrestricted policy. LAADAN-AC removes those actions before selection. Together, the experiments support a narrow interpretation: LAADAN-AC is an admissibility-aware offline policy-learning framework in which

hard masking controls feasible action support and the remaining terms shape behaviour within that support. The reported confidence intervals describe variation across five training seeds only; they do not measure patient-level uncertainty or establish prospective clinical safety.

3.3 Limitations and Future Directions

The present evidence remains benchmark-based. ICU-Sepsis is the primary evaluation environment, and the constructed eICU-CRD Demo Markov Decision Process (MDP) is an exploratory cross-source portability study based on the eICU resource [32], [33]. It does not test clinical deployment, clinician acceptance, prospective intervention, or patient outcomes. Larger evaluations should therefore use independently constructed cohorts such as Medical Information Mart for Intensive Care IV (MIMIC-IV) [34], with separately specified state representations, action spaces, expert references, and admissibility rules. The current admissibility relation is deterministic and benchmark-defined. It does not represent disagreement between clinicians, uncertainty in the constraint itself, delayed contraindications, or treatment risk that depends on the preceding trajectory. The smoothness term is likewise a state-action proxy rather than a temporal regulariser over successive treatment decisions. More fundamentally, the present Lagrangian component constrains an expected benchmark cost and has little independent effect once the hard mask has removed all encoded inadmissible selected actions. Expected-cost control need not constrain the upper tail of cumulative risk [35], while state-constrained Lagrangian formulations show that the appropriate constraint pressure may depend on the current state and uncertain dynamics rather than on a single fixed scalar penalty [36]. We therefore do not interpret the current Lagrangian term as an independent safety guarantee.

LAADAN-AC is also model-free: it does not learn an explicit predictive model of patient dynamics. Consequently, it cannot directly evaluate counterfactual physiological trajectories, propagate uncertainty over future treatment consequences, or distinguish a locally admissible action from one whose delayed consequences may become undesirable. These limitations motivate a predictive constrained-control formulation in which admissibility depends not only on the present action but also on uncertainty-aware estimates of its downstream effects.

We see the natural extension of this work in moving from benchmark-defined action admissibility toward predictive, uncertainty-aware constrained decision-making. Rather than treating the Lagrangian cost as a fixed scalar penalty, we would derive future costs from predicted physiological trajectories, clinical constraint sets, and uncertainty in learned dynamics. Such constraints could depend on state and history, with the dual variable responding to proximity to a risk boundary rather than only to average cost. Distributional risk control, state-dependent Lagrangian formulations, and uncertainty-aware reachability certificates support this direction [35]–[37]. Our aim is not to describe a learned controller as intrinsically safe, but to specify which constraints are enforced, under which assumptions, and how model uncertainty limits those guarantees. A learned world model provides a natural basis for this extension. World-model reinforcement learning has shown that predictive dynamics can support decision-making through imagined futures [38], while causal structured world models have been studied in offline

reinforcement learning to improve generalisation beyond one-step prediction [39]. We would extend LAADAN-AC by learning latent patient dynamics from longitudinal records and evaluating candidate interventions through predicted physiological trajectories. The admissibility interface would remain separate from the predictive model: the world model would estimate future states, rewards, costs, and uncertainty, whereas a constrained decision layer would determine whether those predictions satisfy the required risk criterion. This separation is essential because model error can undermine constraint reasoning; uncertainty-aware reachability [37] and safety-aware planning with learned world models [40] provide complementary directions.

This direction is relevant to medical decision modelling. Model-based reinforcement learning has been explored for intensive-care ventilation, where learned dynamics support reasoning about patient–ventilator behaviour [41], and recent medical world-model work has modelled treatment-conditioned disease evolution and treatment selection [42]. We would evaluate future variants on full eICU and MIMIC-IV resources using independently specified state and action spaces, hospital-level and temporal hold-outs, distribution-shift analysis, stronger off-policy evaluation, and clinically defined uncertainty and admissibility criteria. We would also test whether the constrained world-model interface transfers beyond sepsis. The OhioT1DM dataset contains continuous glucose monitoring, insulin, physiological-sensor, and self-reported life-event measurements for people with type 1 diabetes [43]. Because OhioT1DM is a blood-glucose prediction dataset rather than an established reinforcement-learning benchmark, its use would require an independently defined state, action, reward, and constraint construction. This makes it a useful setting for studying whether admissibility-aware predictive control can generalise from discrete sepsis treatment to continuous or hybrid treatment decisions.

Efficiency should remain part of the research question. Model-based methods may improve sample efficiency, yet learned dynamics and imagined rollouts can increase computational cost. We would report wall-clock time, accelerator-hours, model-update cost, imagined-transition budget, and off-policy-evaluation cost alongside policy performance. Selective imagination, compact latent dynamics, fixed stopping criteria, and reuse of a learned world model across policy updates offer routes to reducing repeated optimisation. This would let us test whether predictive constraint reasoning improves reliability while remaining computationally tractable rather than assuming that a world model is inherently cheaper. Tang and Wiens show that healthcare offline-reinforcement-learning evaluation can itself be computationally demanding [27]. We would also test scalability beyond compact tabular MDPs by increasing decision horizon, state and action complexity, multimodal observation dimensionality, partial observability, stochasticity, and cohort size, while tracking memory use and inference latency to determine whether the constrained predictive interface remains tractable in real-world clinical settings.

4. SUMMARY AND CONCLUSIONS

We introduced LAADAN-AC, an offline actor-critic framework that places benchmark-defined admissibility inside the policy interface and separates feasible action selection from policy shaping. The study uses a predetermined 1,000-epoch protocol, five fixed random seeds, frozen

hyperparameters, and exact finite-horizon evaluation at the final epoch. On ICU-Sepsis, LAADAN-AC achieves 0.7927 ± 0.0012 expected return, zero selected-action inadmissibility, and 0.9540 ± 0.0032 expert argmax agreement. The comparison with BC, CQL-regularised fitted-Q, and unrestricted VOAC shows why these metrics must be considered together: conservative value learning can reduce inadmissible selection, while an unrestricted actor can still leave the benchmark-defined feasible set.

The post-hoc masking experiment provides a second result. Applying the same admissibility mask to final VOAC checkpoints removes inadmissible selections, but the resulting policy retains lower return, weaker expert agreement, larger KL divergence, and greater action deviation than LAADAN-AC. We therefore find that admissibility at action extraction and admissibility during optimisation produce different policies. The ablation study identifies the source of this difference. Removing the action mask reintroduces inadmissible selection, while masking-only and the variant without conservative critic regularisation preserve zero inadmissibility but move to higher-return, lower-alignment operating points. Hard masking therefore determines feasible selected-action support; conservative, expert-guided, smoothness, and Lagrangian terms influence preferences within that support. The state-, action-, value-, and trajectory-level diagnostics show the same mechanism at finer resolution. On the exploratory eICU construction, LAADAN-AC retains zero inadmissibility and 0.9984 ± 0.0006 expert agreement, while CQL attains a slightly higher mean return. This cross-source result preserves the same separation between return, feasibility, and alignment under a different benchmark MDP. Our experiments separate two questions that are often merged in offline treatment-policy learning: which action has high estimated return, and which actions the policy is allowed to select. LAADAN-AC addresses the second question explicitly at the actor interface, then optimises return and expert alignment within the remaining action set. Under the fixed camera-ready protocol, this design produces zero benchmark-defined selected-action inadmissibility while retaining useful return and strong expert agreement. The post-hoc, ablation, portability, and diagnostic studies explain how that behaviour arises and where its limits begin.

We therefore present LAADAN-AC as a reproducible framework for studying admissibility-aware offline policy learning. The future step is to test the same principle under independently defined clinical environments, uncertain and history-dependent constraints, richer patient representations, and prospective decision-support requirements.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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APPENDIX

All supplementary experiments follow the fixed-schedule protocol used for the main results. We evaluate the predetermined epoch-1000 checkpoints from five independently trained seeds (42–46), without selecting checkpoints or adapting hyperparameters using the benchmark evaluator. Reported intervals are 95% t -based confidence intervals across training seeds and therefore quantify variation across training initialisations.

APPENDIX A. POLICY-DISTRIBUTION AND POST-HOC DIAGNOSTICS

A.1. POST-HOC INTERVENTION

What the final mask changes. The paired comparison in Fig. 2 establishes that training under the admissibility interface and masking an unrestricted policy after training produce different final policies. We examine the intervention itself in Fig. A1. For each of the five epoch-1000 VOAC checkpoints, we apply the same benchmark mask used by LAADAN-AC at final action extraction, without retraining the policy. The mask changes VOAC's greedy action in $42.30\% \pm 0.17\%$ of ICU-Sepsis states. Before filtering, the same policies assign $86.32\% \pm 0.15\%$ of their probability mass, on average, to inadmissible actions. Thus, zero inadmissibility after post-hoc filtering does not mean that the unrestricted actor has learned the same action distribution as an actor optimised within the admissible set; in this diagnostic, the final filter performs substantial intervention on the learned policy.

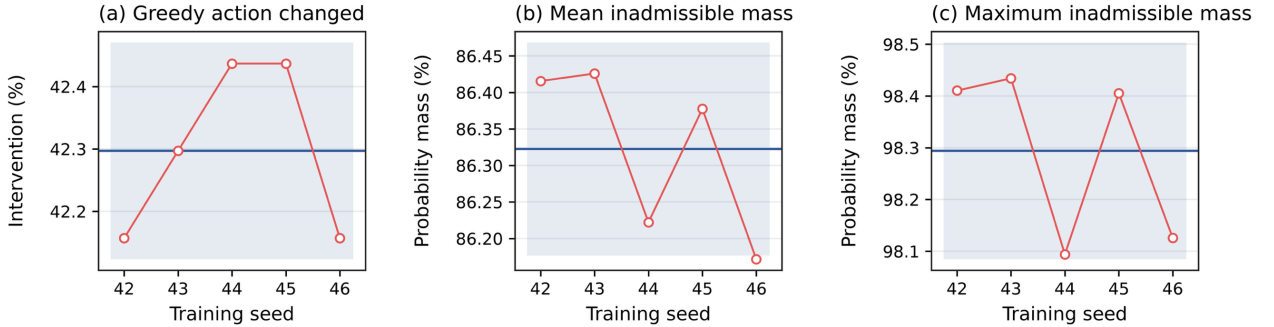


Fig. A1: Post-hoc intervention diagnostic on ICU-Sepsis. Red curves and markers show the five independently trained final VOAC checkpoints. The horizontal blue line denotes the five-seed mean and the shaded region the corresponding 95% t -based confidence interval. The panels report the fraction of states whose greedy action changes after masking, the mean inadmissible probability mass before masking, and the maximum state-level inadmissible probability mass before masking

A.2. DISTRIBUTION-LEVEL ADMISSIBILITY

Feasibility before action extraction. Selected-action inadmissibility describes the final greedy decision, but it does not determine how probability is distributed across the remaining actions. We therefore inspect soft inadmissible probability mass before greedy extraction for the actor-based policies. On ICU-Sepsis, BC assigns $0.669\% \pm 0.010\%$ of its probability mass to inadmissible actions and VOAC assigns $84.622\% \pm 0.150\%$, whereas LAADAN-AC assigns 0.000% . The same ordering appears on the eICU-CRD Demo MDP: BC assigns $0.924\% \pm 0.073\%$, VOAC $78.318\% \pm 0.360\%$, and LAADAN-AC 0.000% . The near-zero view in Fig. A2 resolves the small BC mass from the exact-zero masked distribution. CQL is omitted from this diagnostic because the implemented CQL-regularised fitted-Q comparator defines action values rather than an actor probability distribution.

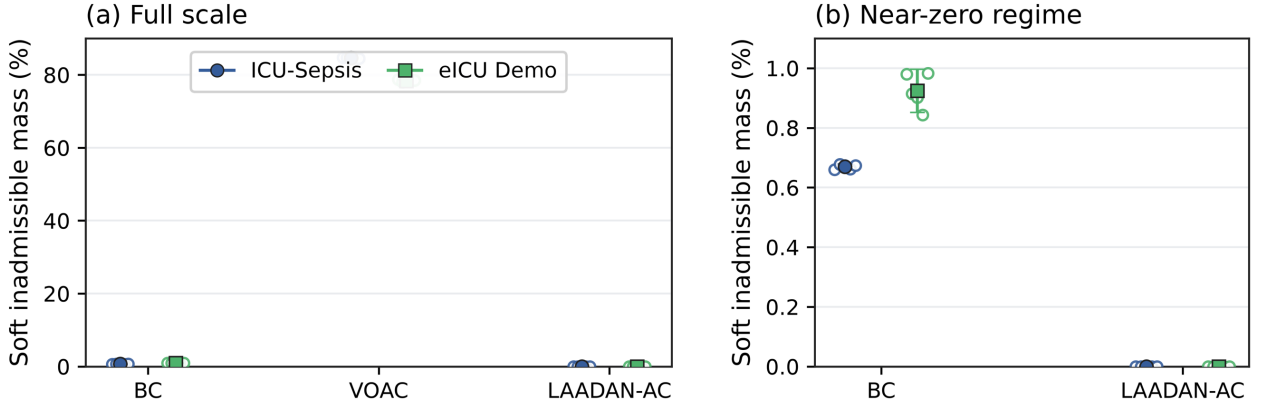


Fig. A2: Distribution-level admissibility at the fixed final checkpoint. The full-scale panel compares soft inadmissible probability mass for BC, VOAC, and LAADAN-AC across ICU-Sepsis and eICU-CRD Demo; the near-zero panel resolves BC and LAADAN-AC at the lower end of the scale. Open markers show individual seeds, filled markers show five-seed means, and error bars show 95% t-based confidence intervals across training seeds

APPENDIX B. OPTIMISATION DYNAMICS UNDER THE FIXED SCHEDULE

Training without benchmark-based checkpoint selection. Fig. B1 reports quantities generated inside the LAADAN-AC optimisation procedure throughout the 1000-epoch training schedule. We show expected training cost, the Lagrange multiplier, expert-policy KL regularisation, and the state-action smoothness term for ICU-Sepsis and eICU-CRD Demo. Benchmark return, selected-action inadmissibility, and expert-agreement values are deliberately absent from these trajectories: under the frozen camera-ready protocol, those evaluation fields remain unset before epoch 1000 and are computed only for the predetermined final checkpoint. The figure therefore characterises the internal optimisation process without reintroducing benchmark performance into model selection.

The two MDPs induce different optimisation trajectories. The expected-cost and regularisation terms settle at different levels across the datasets, while the Lagrange multiplier remains small and evolves gradually over the fixed schedule. We use these traces as optimisation diagnostics only; they are not checkpoint-selection curves and are not evidence of patient-level or clinical convergence.

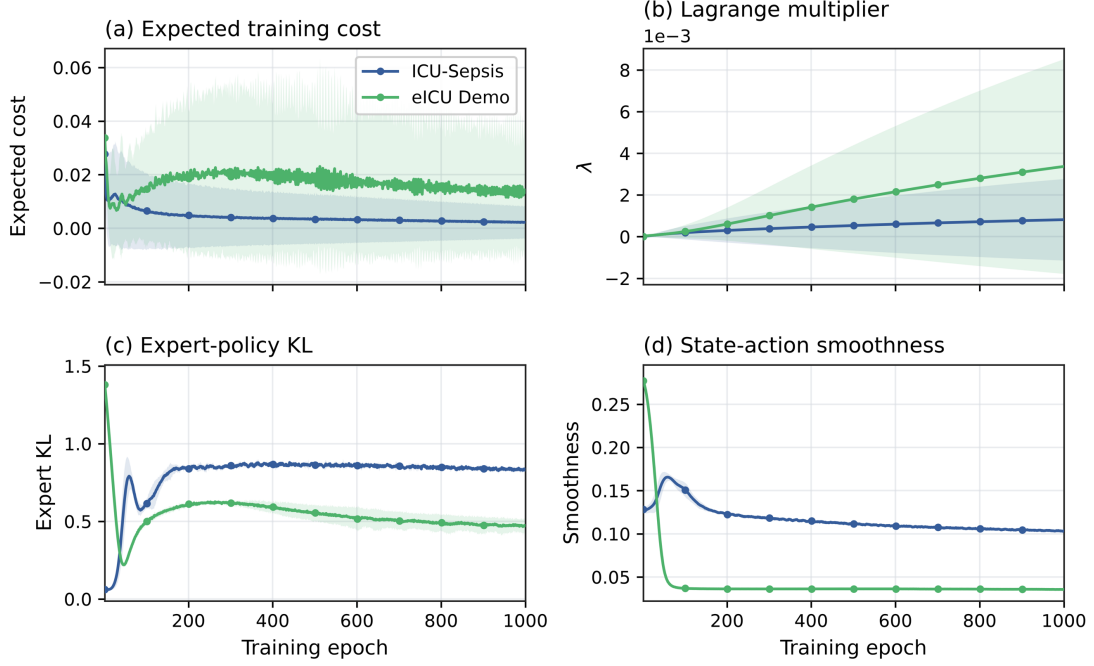


Fig. B1: LAADAN-AC training-objective dynamics under the fixed schedule. Curves report five-seed means over epochs 1–1000 and shaded regions show 95% t-based confidence intervals across training seeds for expected cost, the Lagrange multiplier, expert-policy KL, and state-action smoothness. Markers are shown at regular epochs for readability; the curves are computed from the complete recorded histories. Benchmark return and admissibility are not evaluated for checkpoint selection during these trajectories. Because the intervals are symmetric seed-level t-intervals, a lower confidence-band boundary may extend below zero even when the underlying recorded quantity is non-negative